

Economics of Energy Applications

REE 403

Course Project

What are the impacts of PV panel degradation rate, Initial system efficiency, interest rate, and PV project location on the LCOE of the Saudi Vision 2030 PV goal?

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Abstract

This study investigates the factors influencing the levelized cost of energy (LCOE) of solar photovoltaic (PV) systems in Saudi Arabia, focusing on the country's ambitious renewable energy targets under Saudi Vision 2030. Key parameters analyzed include initial system efficiency, degradation rate, interest rates, and PV project locations. The study shows that higher initial efficiency and lower degradation rates significantly reduce LCOE, while higher interest rates tend to increase energy costs. The results highlight the importance of technological advances, such as improving PV panel efficiency and extending system lifetimes, as well as favorable financial conditions, in making solar energy more economically competitive. This work contributes to continuous initiatives to diversify the energy mixture of Saudi Arabia, provides valuable information on economic execution possibilities of large -scale solar projects, and ultimately a renewable energy source. We support the country. Future research can study more individual solutions based on local conditions and study the role of energy accumulation and policy stimulation in reducing LCOE.

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Chapter One

Introduction

The global energy sector is witnessing a significant transition towards sustainable and renewable energy sources, driven by the urgent need to combat climate change, reduce greenhouse gas emissions, and address the growing global demand for energy. In this context, Saudi Arabia, one of the world's largest producers of fossil fuels, has taken a strategic approach to diversifying its energy portfolio through its ambitious Vision 2030 initiative. This vision aims to reduce the Kingdom's dependence on oil by integrating renewable energy sources, with solar energy playing a pivotal role in achieving this transformation. Saudi Arabia's geographical location within the Sun Belt, offering some of the highest levels of solar irradiance in the world, makes solar photovoltaic (PV) systems an ideal solution for meeting the Kingdom's energy diversification goals.

However, the successful implementation of large-scale solar PV projects in Saudi Arabia comes with unique challenges. The harsh desert environment, characterized by high temperatures, intense sunlight, and frequent dust and sandstorms, can significantly impact the performance, efficiency, and lifespan of PV systems. These environmental factors not only accelerate panel degradation but also necessitate the adoption of advanced technologies and innovative solutions to maintain the efficiency and economic viability of solar projects over time.

A critical metric for assessing the feasibility of solar PV systems is the Levelized Cost of Energy (LCOE), which represents the average cost of electricity generated by a system over its lifetime. Achieving a competitive LCOE is vital for ensuring the long-term sustainability of solar projects. Several factors influence the LCOE of PV systems, including panel degradation rates, initial system efficiency, interest rates, and project location. Each of these factors plays a crucial role in determining the overall cost and energy output of a PV project. For example, lower degradation rates and higher initial system efficiencies can improve energy production and reduce costs, while favorable interest rates and strategic site selection can further enhance project economics.

The Saudi government has introduced several supportive policies and financial incentives, such as subsidies and low-interest financing, to encourage investment in solar energy and accelerate the adoption of renewable energy technologies. These initiatives are expected to play a significant role in overcoming financial and technical barriers to solar deployment. Furthermore, Saudi Arabia's focus on developing regions like Neom and Al Jouf as hubs for renewable energy demonstrates the Kingdom's commitment to leveraging its natural resources to achieve its Vision 2030 goals. This project aims to explore the factors influencing the LCOE of PV systems in Saudi Arabia and provide insights into how these factors can be optimized to ensure the economic viability of solar energy projects. By addressing challenges related to environmental conditions, system performance, and financing, this research seeks to contribute to the broader objectives of Vision 2030, which include positioning Saudi Arabia as a global leader in renewable energy, supporting sustainable development, and contributing to global efforts to transition to clean energy sources.

Research background

Saudi Arabia's Vision 2030 is an innovative strategy aimed at broadening the country's energy sources, with a specific focus on solar energy within its renewable energy plan. Ensuring a competitive Levelized Cost of Energy (LCOE) is crucial for the economic viability of photovoltaic (PV) projects in this perspective. Various elements have a significant impact on the LCOE such as the degradation rate of PV panels, the efficiency of the system at the beginning, interest rates, and the location of the project. The decline in energy conversion efficiency over time, known as the degradation rate of PV panels, directly affects energy output and project economics. Research suggests that the degradation rates usually fall within the range of 0.5% to 1% per year; nevertheless, in Saudi Arabia's severe desert weather, marked by intense heat and dust, these rates could increase unless robust, climate-appropriate PV technologies are utilized (Jordan & Kurtz, 2013). Moreover, the efficiency of the system at the beginning, which depends on the quality of PV modules, inverters, and overall system design, is crucial in lowering LCOE. Systems with increased efficiency produce greater energy output per given area, leading to substantial cost reductions. With Saudi Arabia's high levels of solar radiation, improving system efficiency presents significant chances to lower costs.

Aside from technical variables, financial factors like interest rates greatly impact the levelized cost of electricity (LCOE). Reduced interest rates decrease capital costs, making project financing more budget-friendly. The Saudi government has made it easier by providing subsidies and favorable financing to encourage private-sector investments in solar energy (Lazard, 2023). Finally, the LCOE of PV projects is impacted by differences in solar resource availability, land costs, and distance to transmission infrastructure based on the geographical location. Saudi Arabia's location in the Sun Belt, especially areas like Neom and Al Jouf, offers significant potential for solar energy. Yet, it is crucial to choose the best site and connect it efficiently to the national grid in order to reduce transmission losses.

Research motivation

The reason for this study stems from the increasing worldwide need for sustainable energy solutions and Saudi Arabia's dedicated focus on renewable energy as outlined in Vision 2030. Being one of the leading oil producers, the Kingdom is looking to expand its energy sources to lessen reliance on fossil fuels and guarantee sustainable energy security. Solar energy is crucial in reaching these objectives due to its vast potential in Saudi Arabia. Nevertheless, in order for solar photovoltaic (PV) systems to become economically feasible and extensively embraced, the Levelized Cost of Energy (LCOE) needs to be reduced. Addressing important factors like PV panel degradation, system efficiency, financing conditions, and ideal project locations is necessary for this.

Saudi Arabia's distinctive weather conditions, such as elevated temperatures, dust, and occasional sandstorms, create substantial obstacles for the effectiveness and durability of PV systems. These environmental conditions speed up panel deterioration, impacting the overall energy production and financial viability of the system. It is essential to comprehend these effects and create plans to reduce them in order to guarantee the prosperity of solar initiatives in the area.

Additionally, the competitiveness of solar energy projects is directly impacted by the initial efficiency of the system and financing conditions. Government-led programs provide subsidies and low-interest financing options, encouraging the exploration of financial models to further decrease the LCOE of solar systems. Furthermore, taking advantage of Saudi Arabia's geographical benefits, like the abundant solar energy in areas like Neom and Al Jouf, could optimize energy generation and reduce expenses.

This study is driven by the necessity to connect technical, environmental, and financial factors in solar PV projects. The study aims to provide practical insights that support Vision 2030's renewable energy goals and the global shift towards sustainable energy systems by examining the interaction of these factors. Ultimately, the goal of this study is to improve the effectiveness and trustworthiness.

Research outline

- Saudi vision 2030 goal
- PV Panel Degradation Rate
- Initial System Efficiency
- Interest Rates
- PV Project Location
- LCOE
-

Chapter Two

Literature Review

Saudi Vision 2030

The Saudi Vision 2030 framework aims to diversify the Kingdom's economy and decrease its dependence on oil by enhancing innovation and sustainability. Renewable energy is a keystone in this plan, with specific targets for solar power-including 40 GW of photovoltaic capacity by 2030. It also aligns with the general global sustainability objectives of reducing carbon emissions and fighting climate change.

Meanwhile, Saudi Arabia has made considerable progress, with multiple solar projects such as

1-Sakaka PV

The Sakaka Solar Power Plant Project in Al Jawf is under the Ministry of Energy and was the first Renewable Energy project to be guided by King Salman bin Abdulaziz Al Saud. The plant uses solar energy through photovoltaic technology (PV) to generate electricity. The Sakaka Solar Power Plant consists of 1.2 million solar panels arranged over 6 km² of land. The site was meticulously selected by a specialized Saudi technical team to ensure the highest possible quality of electrical power production. Contributing to the reduction of 606,000 tons of carbon emissions. The plant's production capacity is 300 MW. 44,000 housing units were covered. The project set a new world record for low costs in the solar PV sector, which amounted to 0.08775 SAR per kilowatt hour[1]



Figure 1 Sakaka PV IPP

2-Sudair Solar PV

located in Saudi Arabia's Riyadh region, is one of the largest and most significant solar initiatives under the Kingdom's Vision 2030 plan. With a capacity of 1.5 GW, Sudair is set to be a key contributor to Saudi Arabia's renewable energy targets, expected to generate enough power to supply electricity to 185,000 homes and offset 2.9 million tons of CO₂ emissions annually. The project has recorded the second lowest Levelized Cost of Electricity (LCOE) of 1.239 cents per kWh [USD 1.239 cents/kwh].[2]



Figure 2 Sudair PV IPP

Layla Solar PV Plant

The Layla Solar PV Plant is a pivotal part of Saudi Arabia's renewable energy expansion under the National Renewable Energy Program (NREP). Located in Riyadh, this project is designed to produce 80 MW of electricity, contributing to the country's goals of reducing dependency on fossil fuels. By utilizing solar photovoltaic technology, it aligns with Vision 2030's aim of developing a sustainable energy infrastructure while minimizing environmental impact[3]



Figure 3 Layla Solar PV IPP

Rabigh Solar Project

Situated in the western region of Saudi Arabia, the Rabigh Solar Project focuses on leveraging the Kingdom's high solar irradiance. While details of its operational status remain under development, this project is an integral part of the national strategy to achieve the 40 GW renewable energy target by 2030. It highlights Saudi Arabia's commitment to transitioning to clean energy while fostering private sector involvement[4]

Solar PV Cell & Module Manufacturing Plant

Saudi Arabia is also investing in localized solar energy technology through the Solar PV Cell & Module Manufacturing Plant. Established in 2010 by [the King Abdulaziz City for Science and Technology](#) (KACST), the Solar PV Cell & Module Manufacturing Plant and PV Reliability Laboratory produces solar panels and cells. The facilities will bring the latest solar energy technologies to Saudi Arabia and create top-quality equipment that can withstand extreme heat and sandstorms.[5]



Figure 4 Solar PV Cell & Module Manufacturing Plant

2.2 PV Panel Degradation Rate

The degradation rate of a photovoltaic (PV) panel refers to the gradual decline in its ability to convert sunlight into electricity over time. This degradation is a natural process caused by environmental exposure, material wear, and operational factors. Typically, modern PV panels degrade at an average rate of 0.5% per year, meaning that after 25 years, most panels retain approximately 87.5% of their original efficiency. Understanding degradation is essential for predicting the performance, financial viability, and sustainability of PV systems over their lifetime. Some companies like SUNPOWER guarantee 98% production for the first year, and degradation of no more than 0.25% per year for the following 24 years. The warranty guarantees that the power output at the end of the 25th year will be at least 92% of the Guaranteed Peak Power rating. [6]

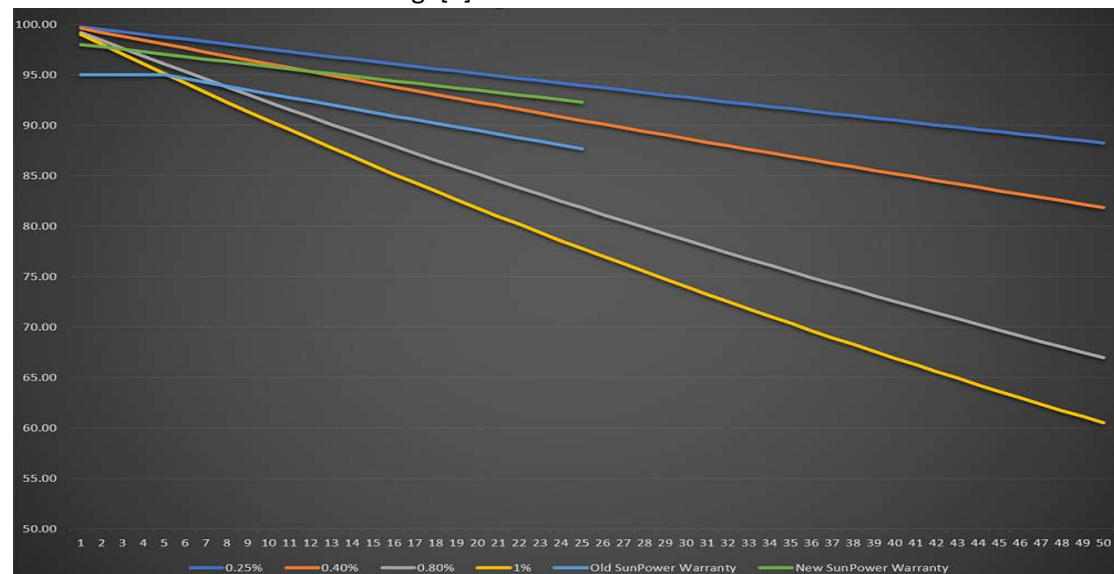


Figure 5 Panel degradation

Causes of Degradation

PV panel degradation occurs due to both intrinsic and extrinsic factors. **Intrinsic degradation** includes material aging and microcrack formation in silicon cells, while **extrinsic factors** include environmental stressors such as UV radiation, thermal cycling, and moisture ingress. These stressors accelerate the breakdown of encapsulants, frame materials, and anti-reflective coatings. Manufacturers combat these issues by employing advanced materials and protective technologies, but no panel is entirely immune to degradation. [7]

PV Panel Degradation

As panels degrade, their ability to produce electricity diminishes, reducing energy output and impacting financial returns. This reduction directly affects metrics like the Levelized Cost of Electricity (LCOE) and Return on Investment (ROI). For example, panels with higher degradation rates may fail to meet production guarantees, requiring early replacement. Accurate modeling of degradation is crucial for long-term project planning and financing. [8]

Importance of Initial Quality

The initial build quality of PV panels plays a significant role in degradation rates. High-quality panels from reputable manufacturers often exhibit lower and more predictable degradation rates due to rigorous testing, better materials, and advanced engineering. Certification standards, such as those from IEC and UL, ensure panels meet reliability benchmarks before deployment.[9]

Climatic and Environmental Conditions

The operating environment significantly influences degradation. In hot, humid, or high-UV regions, panels are exposed to more severe stress, leading to faster degradation. For instance, desert climates can cause increased wear on anti-reflective coatings and encapsulants. Selecting climate-optimized panels and employing protective measures like proper mounting and cooling systems can mitigate these effects. The biggest cause of PV panel degradation in Saudi Arabia is **dust accumulation** due to frequent dust storms and the high dust intensity in desert environments. Dust deposition reduces sunlight reaching the panels, leading to efficiency losses of up to 20-40% if not cleaned regularly. Additionally, **high temperatures** accelerate thermal degradation and material aging, while **UV exposure** contributes to encapsulant and anti-reflective coating wear. These factors collectively lead to higher degradation rates compared to more temperate climates[10]

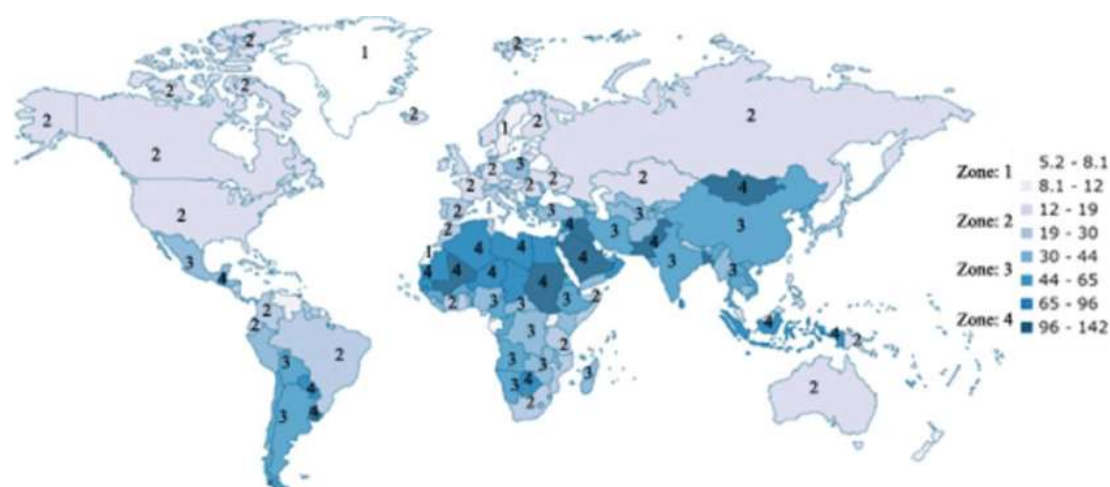


Figure 6 Dust intensity around the world

Advances in PV Technology to Reduce Degradation

Innovations in PV panel technology aim to minimize degradation rates. Improvements include the use of monocrystalline over polycrystalline silicon, better encapsulation materials, and advanced coatings that resist environmental damage. Some manufacturers now offer panels with degradation rates as low as 0.2% per year, extending the productive life of installations beyond 30 years.[11]

2.3 initial system efficiency

The initial system efficiency of a PV system plays a significant role in determining how degradation will affect its long-term performance. A system with higher initial efficiency typically has a higher power output, which makes it more resilient to the impacts of degradation over time. As degradation mechanisms, like **Light-Induced Degradation (LID)** and **Potential-Induced Degradation (PID)**, cause a decrease in panel performance, systems with a higher starting efficiency maintain better overall output for a longer period[11]

Several factors related to **initial system efficiency** impact degradation:

Panel Type and Quality

Panel type and quality are crucial elements influencing the **initial system efficiency** of a photovoltaic (PV) system and how it performs over time. Different types of panels, such as **monocrystalline**, **polycrystalline**, and **thin-film**, exhibit varied levels of efficiency and degradation behavior, which directly affect the long-term performance of a PV system.[12]

Monocrystalline Panels: Known for their high efficiency, monocrystalline silicon panels typically have higher initial system efficiency compared to other types. Their efficiency typically ranges from 18% to 22%, and they generally experience lower degradation rates over time. The higher purity silicon used in monocrystalline panels ensures better performance, even as degradation factors like **light-induced degradation (LID)** and **potential-induced degradation (PID)**, these panels tend to perform better under high temperatures and low-light conditions, further extending their efficiency over their lifespan[11]

Polycrystalline Panels: Polycrystalline panels are generally less efficient than monocrystalline panels, with initial efficiencies between 15% and 18%. While they are more affordable, they also tend to have a higher rate of degradation compared to monocrystalline panels, particularly when exposed to high temperatures or prolonged UV radiation. Over time, this higher degradation rate leads to a greater loss in performance[11]

Thin-film : Thin-film technology, including **amorphous silicon** and **cadmium telluride (CdTe)**, offers lower initial efficiency (around 10% to 12%) but can be beneficial in certain applications due to their flexibility and lower production costs. However, they typically degrade faster due to their susceptibility to factors like **moisture ingress** and **UV radiation**. Thin-film panels may also experience **light-induced degradation** more severely in their early years of operation, leading to a quicker decline in energy output[11]

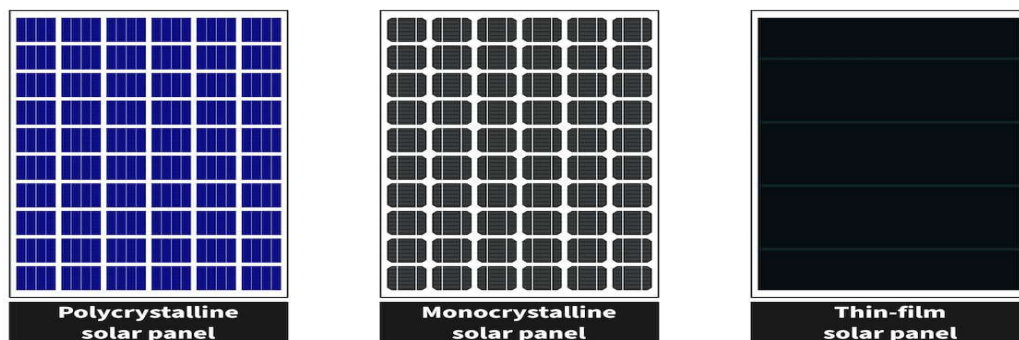


Figure 7 Different types of panels

Manufacturing Quality

Manufacturing quality plays a significant role in determining the **initial system efficiency** of a photovoltaic (PV) system. The precision and methods used in manufacturing directly affect the panel's performance, lifespan, and its degradation over time. High-quality panels often exhibit lower degradation rates and maintain their efficiency for longer periods.

- A. Cell Efficiency:** The manufacturing process affects the efficiency of the solar cells. Advanced technologies like **Passivated Emitter and Rear Contact (PERC)** and **n-type silicon** cells enhance the initial efficiency of PV panels by reducing energy losses caused by surface recombination. These innovations result in higher energy conversion rates, especially in monocrystalline panels, and mitigate performance drops over time[13]

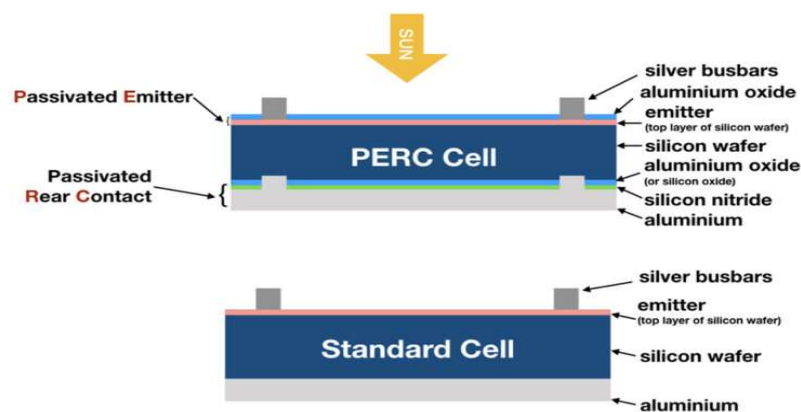


Figure 8 PERC Cell

- B. Encapsulation and Coating Quality:** The quality of encapsulation materials (such as EVA or PVB) and protective coatings plays a crucial role in preventing moisture ingress, UV degradation, and thermal stress. Panels manufactured with superior materials are more resistant to delamination and microcracks, which are common causes of degradation in lower-quality panels[14]

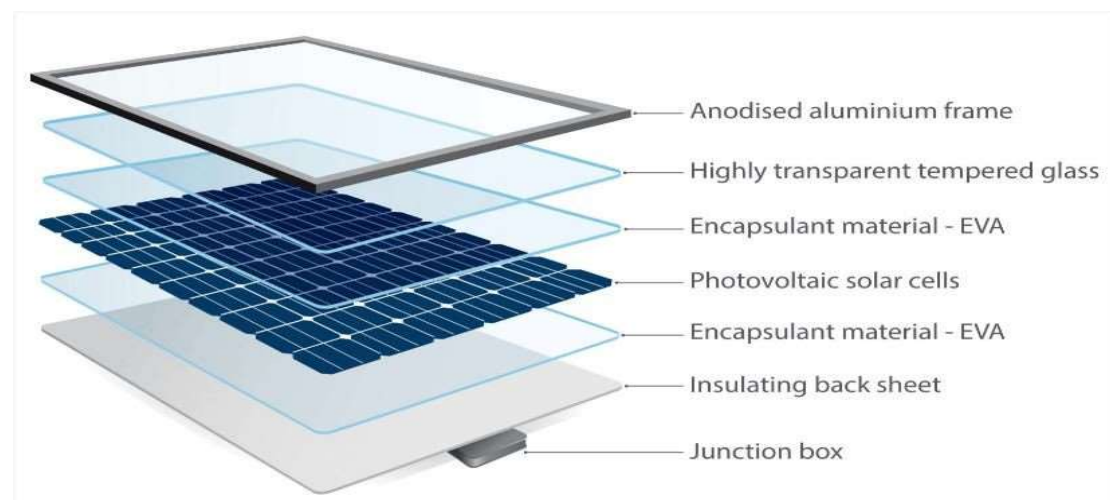


Figure 9 EVA in Solar cell

- C. Manufacturing Standards and Certification:** Panels produced under strict **international standards** (such as IEC 61215 or UL 1703) are tested for performance, durability, and safety. Panels with higher manufacturing standards are less prone to defects and fail less often in real-world conditions, leading to lower initial degradation rates. Furthermore, panels that undergo rigorous quality control measures during production tend to exhibit higher consistency in performance[15]
- D. Material Purity:** The use of high-purity silicon and other high-quality materials results in panels with fewer defects in the crystal structure, leading to better energy output and less degradation over time. Lower-quality materials can lead to a higher incidence of **microcracks**, which gradually reduce the panel's efficiency[7]

Temperature Coefficients

Temperature are a critical factor that affects the initial system efficiency of photovoltaic (PV) panels. The temperature coefficient of a solar panel indicates how much its efficiency decreases as the temperature increases. This is particularly important for solar panels used in hot climates, where high temperatures can significantly reduce performance over time. it quantifies the decrease in power output (in percentage terms) for every degree Celsius increase in temperature above 25°C (the standard test condition). For example, a typical temperature coefficient for monocrystalline panels ranges from -0.3% to -0.4% per degree Celsius, while polycrystalline panels often have a coefficient around -0.4% to -0.5% per degree Celsius[16]

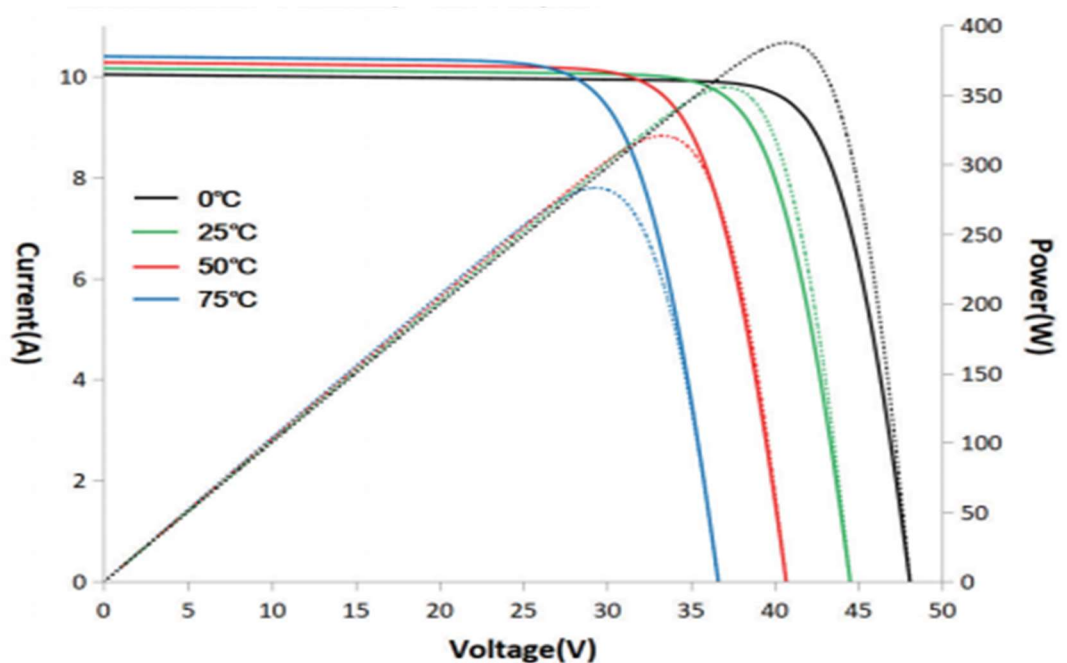


Figure 10 I-V and P-V characteristics of solar cells

Design and Installation

The design and installation of photovoltaic (PV) systems significantly impact their initial efficiency. Proper panel orientation and tilt are crucial for maximizing sunlight capture, while avoiding shading from trees or buildings prevents power loss. Adequate spacing between panels ensures proper air circulation to prevent overheating, which would otherwise reduce efficiency due to higher temperature coefficients. Additionally, the quality of wiring and inverter systems directly affects energy conversion, and professional installation helps avoid issues like microcracks and electrical faults. Overall, optimized design and careful installation can enhance performance and longevity, reducing the risk of early degradation. [17]

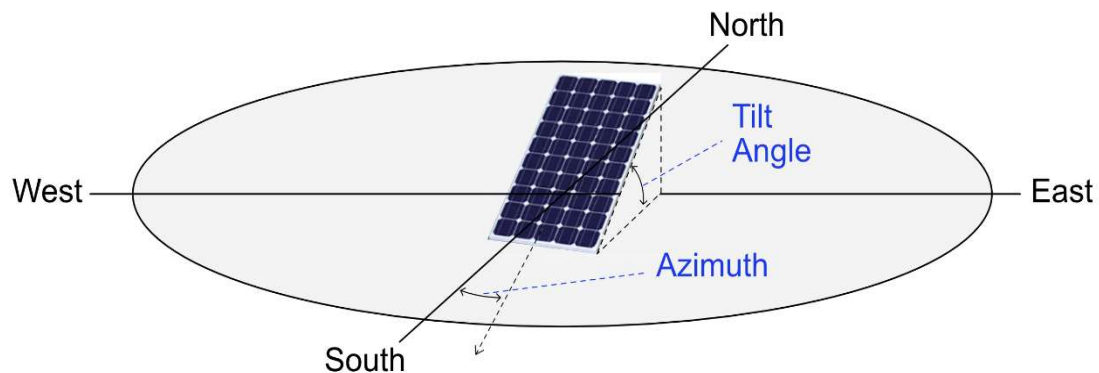


Figure 11 Solar angles

2.4 interest rate

interest rate is the cost of borrowing money or the return on investments, expressed as a percentage of the principal amount over time, and it plays a significant role in the financing and economic viability of photovoltaic (PV) projects. In the context of PV systems, interest rates determine the cost of capital for developers, influencing upfront investment requirements and long-term project cash flows. Lower interest rates reduce financing costs, enabling more competitive energy pricing and improving the attractiveness of solar projects. Conversely, high-interest rates increase the Levelized Cost of Energy (LCOE) by raising the expense associated with loans or other forms of debt financing. Interest rates also affect the discount rate used in LCOE calculations, impacting the valuation of future energy production and operational expenses. Thus, managing interest rates through favorable financing structures, such as green bonds or subsidized loans, is critical for achieving cost-efficient solar energy deployment[18]



Figure 12 Saudi Arabia's Short Term Interest Rate from Jan 2007 to Aug 2024 in the chart

2.5 PV location

The location of a photovoltaic (PV) system is a decisive factor in its performance and economic viability. Solar irradiance, which varies by location, is the primary driver of energy output. Areas with high solar irradiance, such as those in Saudi Arabia, benefit from increased electricity generation potential, leading to lower levelized costs of electricity (LCOE). However, location-specific challenges such as extreme temperatures and environmental conditions must also be considered. [19] In desert regions, for instance, high ambient temperatures can negatively impact PV efficiency due to thermal losses. A study conducted by KAUST in Saudi Arabia highlighted that modules operating in high-temperature environments regularly reach 60–65°C, reducing power output and potentially shortening their lifespan

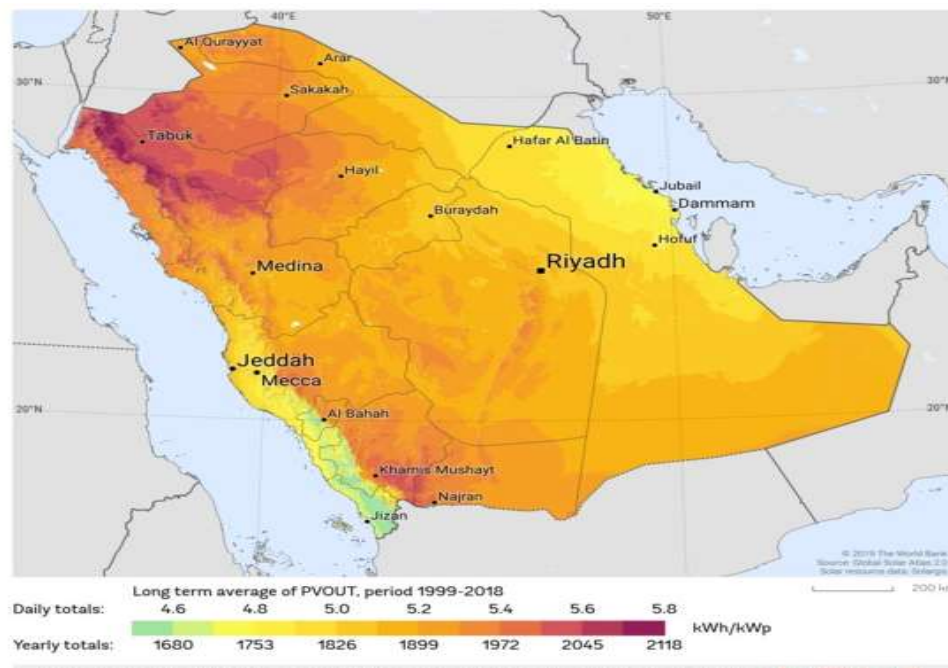


Figure 13 Photovoltaic Electricity Potential in Saudi arabia

2.6 Levelized Cost of Electricity (LCOE)

The Levelized Cost of Electricity (LCOE) is a widely utilized metric to assess the economic viability of energy-generating technologies, including photovoltaic (PV) systems. LCOE represents the average cost per kilowatt-hour (kWh) of electricity over the lifetime of an energy system, accounting for all costs, including initial capital, operations and maintenance (O&M), and end-of-life costs, discounted to present value. Studies show a significant decline in PV LCOE globally, driven by advances in technology, economies of scale, and supportive policies. For instance, between 2010 and 2020, global average PV LCOE dropped by 82%, with further reductions in regions like Saudi Arabia, where abundant solar resources and competitive bidding under Vision 2030 yielded record-low costs. Key factors influencing LCOE include degradation rates, which affect long-term performance, and discount rates, which vary by region and impact financing costs. Additionally, assumptions about system lifetime, such as operational spans of 20 versus 30 years, significantly alter LCOE outcomes. [20]

Chapter Three

Research Methodology

In This section present equations that will be used to assess the PV project

To calculate LCOE

$$LCOE = \frac{I_0 + \sum_{t=0}^n O\&M_t}{\sum_{t=0}^n E_t}$$

where I_0 is the initial investment, $O\&M_t$ is the operations and maintenance costs during a single period t , n is the number of time periods that correspond to the project's lifetime, and E_t is the total energy generated during a single period t .

However, this equation does not consider the interest rate on the initial investment, so we adjusted the equation to take interest rate into consideration

$$LCOE = \frac{I_0 + \text{interest cost} + \sum O\&M}{\sum E_t}$$

Where

$E = \text{Solar System Power Rating} \times \text{Average Peak Sun Hours per Day} \times 365 \times \eta \times (1 - \text{Deg.R})^t$

Since we are going to study the impacts of PV panel degradation rate, Initial system efficiency, interest rate, and PV project location on the LCOE of the Saudi Vision 2030 PV goal we are going to depend on available data from completed solar projects in Saudi Arabia

Table 1

Project	Capacity	Initial Investment Cost (\$/W)
Sakaka PV	300 MW	\$1.01/W
Al-Shuaibah	2660 MW	\$0.91/W
Sudair PV	1500 MW	\$0.60/W
Layla Solar PV	91 MW	Approx. \$0.64/W

[1], [2], [3]

We will consider the average \$0.79/W as initial investment cost in the calculations to conduct the research. Solar system power rating will be considered 40 GW as Saudi Arabia aims to achieve in 2030 vision. GHI in Saudi Arabia varies by region, with annual averages typically ranging between 5.5 to 6.5 peak sun hours per day so we will consider 6 hours per day for the calculations. Operation and maintenance (O&M) costs for photovoltaic (PV) projects in Saudi Arabia are relatively low, reflecting the country's favorable solar conditions and the efficiency of its utility-scale solar plants . While it can vary based on project size, technology, and location, industry estimates suggest that O&M costs for large-scale PV plants in the region average around \$10 per kilowatt (kW) per year, and that is what we are going to assume in the calculations. The selling price of electricity in Saudi Arabia varies depending on the customer category (residential, commercial, industrial, or governmental) and the level of consumption.

Chapter Four

Results & Discussion

4.1 Interest rate =0, Initial efficiency =1 and varying Degradation rate

This chart illustrates that the LCOE decreases as the initial system efficiency increases. Higher efficiency systems generate more electricity for the same installed capacity, reducing the cost per kilowatt-hour of energy produced. This highlights the importance of optimizing system efficiency to improve the economic viability of solar projects by maximizing energy output while minimizing costs.

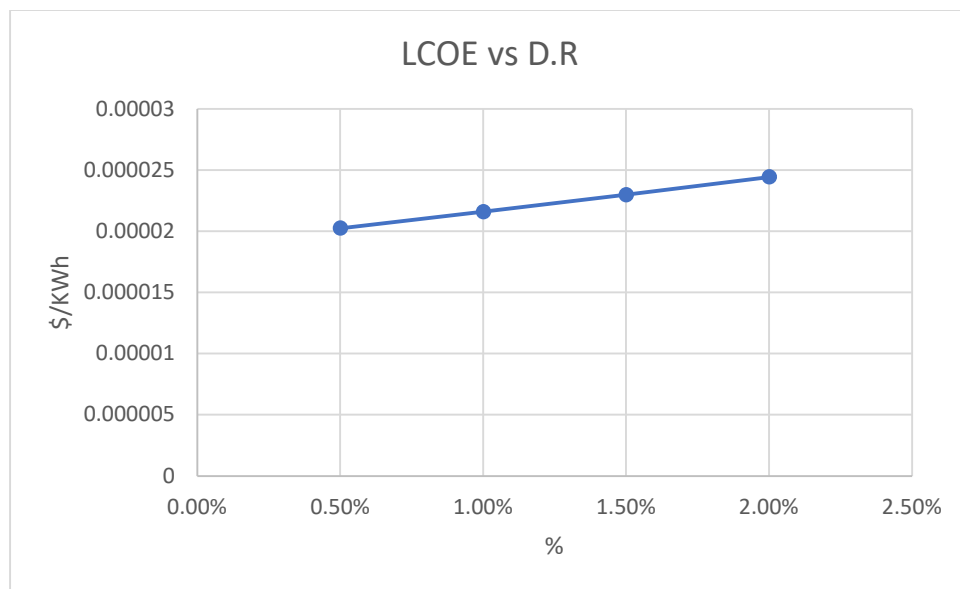


Figure 14 LCOE vs D.R

4.2 Degradation rate =0, Initial efficiency =1 and varying interest rate

The chart shows that the LCOE rises as the interest rate increases. Higher interest rates result in greater financing costs for the project, which directly impacts the overall cost of electricity generation. This emphasizes the critical role of securing low-interest loans to reduce the financial burden on the project and enhance its cost-effectiveness over its lifetime.

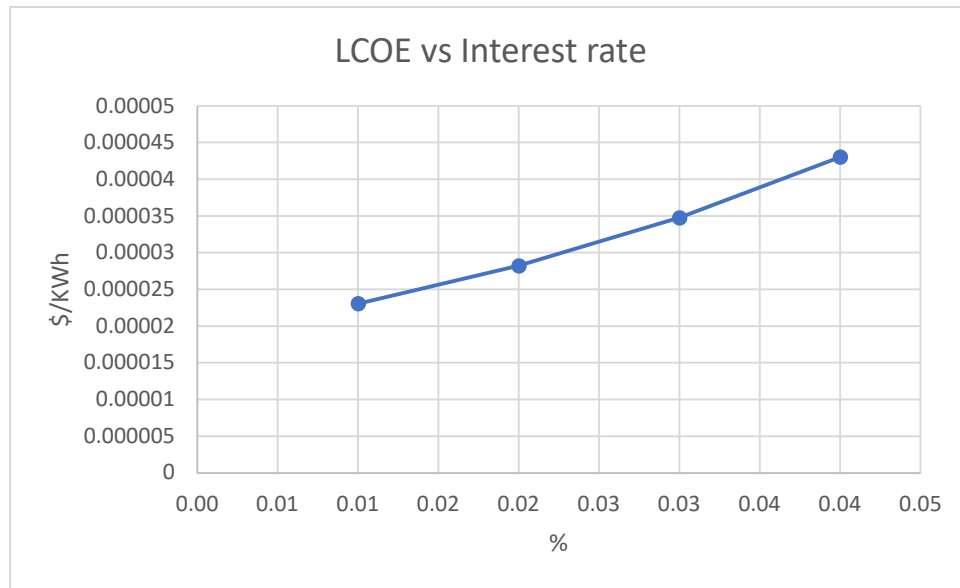


Figure 15 LCOE vs Interest rate

4.3 Degradation rate =0, Interest rate =0 and varying Initial efficiency

This chart demonstrates that the LCOE increases with a higher degradation rate. Systems with higher degradation rates lose their energy production capacity more quickly over time, leading to reduced total energy generation throughout their lifespan. As a result, the cost of electricity per kilowatt-hour rises, underlining the importance of using high-quality materials and proper maintenance to minimize system degradation and maintain long-term performance.

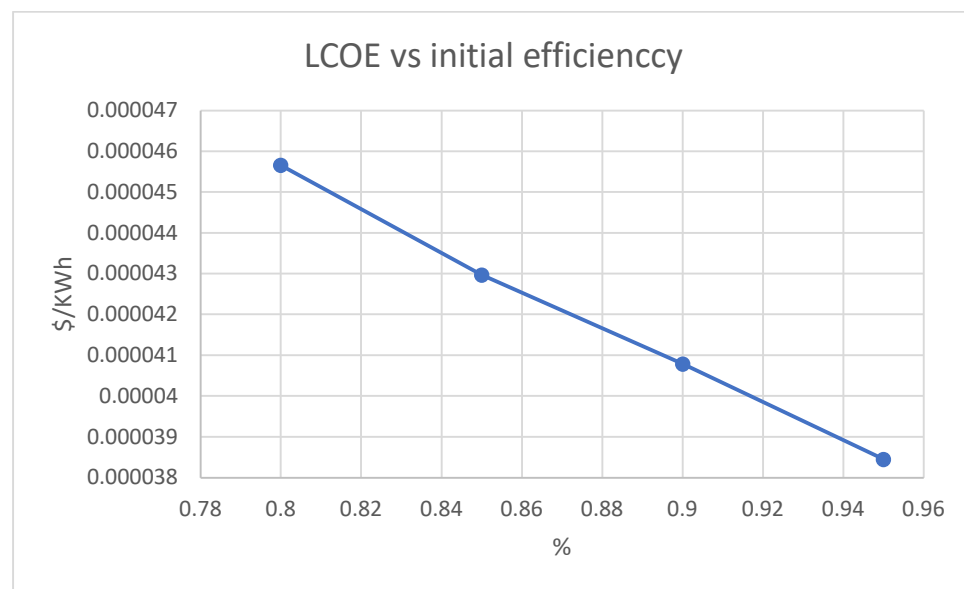


Figure 16 LCOE vs Initial Efficiency

Discussion

The findings show that the Levelized Cost of Energy (LCOE) is greatly affected by key system factors such as starting efficiency, interest rate, and degradation rate. Higher initial system efficiency leads to reduced LCOE due to increased electricity generation for the same capacity, allowing for better cost distribution over a higher energy output. This enhancement underscores the significance of using high-performance PV panels and advanced system designs to improve overall effectiveness. Conversely, when the interest rate goes up, the LCOE also goes up considerably because the increased financing expenses lead to a rise in the overall project costs. This highlights the importance of obtaining low-interest loans or incentives to ensure the economic viability of solar projects. Furthermore, the speed at which degradation occurs is crucial, as a faster rate will hasten the decrease in the system's energy generation over time, ultimately decreasing the total energy output over its lifetime and raising the LCOE. These results highlight the importance of giving precedence to advanced materials and maintenance strategies in order to reduce degradation. In general, improving efficiency, obtaining good financing terms, and reducing degradation are important in reaching the lowest LCOE for cost-effective and sustainable solar energy deployment.

Chapter Five

Conclusion & Future Work

In conclusion, this project uncovered the key factors that influence the levelized cost of electricity (LCOE) of a photovoltaic (PV) system and provided a comprehensive analysis of how variables such as initial system efficiency, degradation rates, and financial parameters such as interest rates affect the economic feasibility of a solar energy project. The results highlight the importance of improving solar panel efficiency and minimizing degradation rates over time to achieve reduced energy costs and improved financial sustainability. Additionally, obtaining favorable financial terms, such as low-interest loans, could significantly reduce the LCOE, making solar energy a more competitive and viable alternative. These ideas contribute to the broader goals of Saudi Vision 2030, which highlights the role of cost-effective, sustainable renewable energy in reducing reliance on fossil fuels and helping to diversify the economy. Looking to the future, there are significant opportunities in this area for future work. Research efforts can be focused on advancing photovoltaic technology to achieve higher efficiencies and lower degradation rates, thereby extending system life and improving overall performance. Integrating energy storage systems with solar PV systems, such as battery technologies, can be explored to improve energy reliability and reduce reliance on the electricity grid. Additionally, further research on the impact of regional climatic conditions, including high temperatures and dust accumulation, on PV system performance could provide solutions more suited to Saudi Arabia's unique environment. Policy research aimed at developing effective incentives such as subsidies, feed-in tariffs, and tax breaks will continue to foster the adoption of renewable energy systems. By addressing these areas, future research can build on this work and accelerate the transition to a sustainable energy future while aligning with Saudi Arabia's ambitious renewable energy goals.

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